Quantifying Logical Consistency in Transformers via Query-Key Alignment

Anonymous ACL submission

Abstract

Large language models (LLMs) have demonstrated impressive performance in various natural language processing tasks, yet their ability to perform multi-step logical reasoning remains an open challenge. Although Chain-of-Thought prompting has improved logical reasoning by enabling models to generate intermediate steps, it lacks mechanisms to assess the coherence of these logical transitions. In this paper, we propose a novel, *lightweight* evaluation strategy for logical reasoning that uses query–key alignments inside transformer attention heads. By computing a single forward pass and extracting a "QK-score" from carefully chosen heads, our method reveals latent representations that reliably separate valid from invalid inferences, offering a scalable alternative to traditional ablation-based techniques. We also provide an empirical validation on multiple logical reasoning benchmarks, demonstrating improved robustness of our evaluation method against distractors and increased reasoning depth. The experiments were conducted on a diverse set of models, ranging from 1.5B to 70B parameters.

1 Introduction

Large language models (LLMs) have achieved remarkable success in a range of NLP tasks, yet they still struggle with reliable multi-step logical reasoning (Wei et al., 2022; Kojima et al., 2022; Yang et al., 2024; Seals and Shalin, 2023; Wan et al., 2024). While chain-of-thought prompting has improved performance by allowing models to generate intermediate reasoning steps, it lacks a mechanism to assess the coherence of these transitions.

In this work, we propose a novel evaluation method that uses certain internal Query-Key (QK) interactions within transformer heads as a proxy for logical consistency. Specifically, given a triplet—context c, statement s, and candidate answer a_i, a_0 = true or a_1 = false —we define the QK-score as:

$$S_{QK}^{(l,h)}(c, s, a_i) = \boldsymbol{q}_{a_i}^{(l,h)\top} \boldsymbol{k}_s^{(l,h)},$$

where $\boldsymbol{q}_{a_i}^{(l,h)}$ is the query vector for the answer token and $\boldsymbol{k}_s^{(l,h)}$ is the key vector corresponding to the statement, see Section 3. Our method efficiently identifies transformer heads capable of accurately evaluating the validity of logical transitions. It processes all heads in a single run, making it scalable to large models.

The contributions of this paper are as follows:

- We propose a novel QK-score mechanism that uses the interactions between certain query- and key-vectors to assess the correctness of logical transitions in the corresponding tasks.
- We conduct extensive experiments on multiple logical benchmarks to demonstrate that our method improves logical consistency, especially in multi-step reasoning scenarios or when context contain reasoning distractors.

2 Related Work

A significant line of research has focused on improving multi-step logical reasoning in LLMs via chain-of-thought (CoT) prompting (Wei et al., 2022; Kojima et al., 2022; Yang et al., 2024; Seals and Shalin, 2023; Wan et al., 2024). Although CoT methods allow models to generate intermediate reasoning steps, they lack a mechanism to assess the coherence of these logical transitions.

Mechanistic interpretability studies have examined the roles of transformer attention heads. Recent work revealed that attention layers operate in phases—knowledge recalling, latent reasoning, and expression preparation (Elhage et al., 2021; Olah et al., 2020; Zheng et al., 2024). Subsequent studies have shown that some attention heads introduce biases by evenly splitting probabilities between

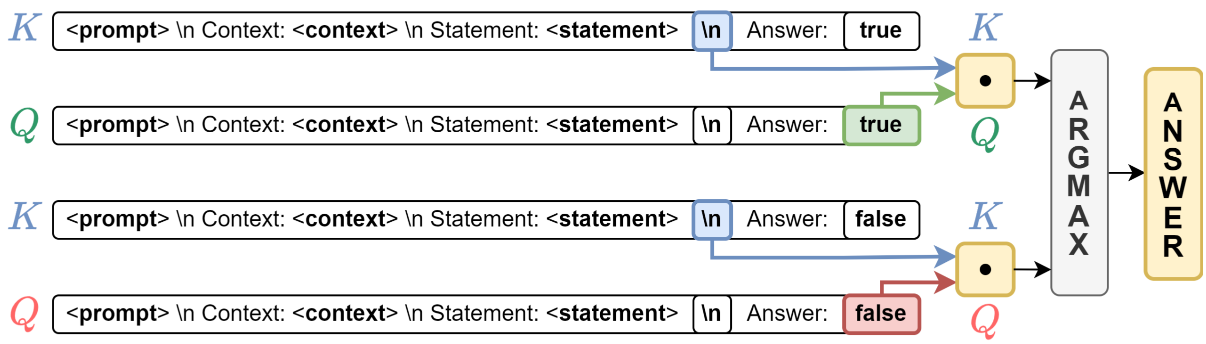

Figure 1: Our method calculates the Query-Key score between the end-of-line token immediately after the statement and the "true"/"false" tokens, for the designated head, from which we derive the answer.

answer options (Lieberum et al., 2023; Yu et al., 2024), while others suppress such behaviors during the final expression phase (Kim et al., 2024). In addition, recent investigations have attempted to analyze model behavior by disabling specific components (Zhang and Nanda, 2024; Todd et al., 2024; Yao et al., 2024), though these approaches are often computationally expensive or limited to simpler tasks (Wang et al., 2023; Yao et al., 2024).

For an expanded discussion on related works, see Appendix A.

3 Approach

Our method evaluates logical consistency by examining certain internal Query-Key (QK) interactions within transformer heads. In our setup, each input consists of a context c (which provides the premises), a statement s (a candidate conclusion), and a candidate answer a_i (with a_0 = true and a_1 = false), see Figures 2, 4, 5 for examples.

For every attention head, indexed by h in layer l, we compute a QK-score that quantifies the alignment between the representation of the statement and that of the candidate answer. Given the triplet (c, s, a_i), the **QK-score** is the dot-product $S_{QK}^{(l,h)}(c, s, a_i) = \boldsymbol{q}_{a_i}^{(l,h)\top} \boldsymbol{k}_s^{(l,h)}$, where $\boldsymbol{q}_{a_i}^{(l,h)}$ is the query vector associated with the token representing a_i (either "true" or "false") and $\boldsymbol{k}_s^{(l,h)}$ is the key vector corresponding to the token marking the end of the statement (see Figure 1). This score reflects how well a head can distinguish valid logical transitions.

By evaluating all heads in a single forward pass, our approach identifies those that reliably assess logical consistency. This efficient procedure avoids the need for extensive model modifications or head-by-head ablation studies that are common in prior work.

Use provided context and answer whether the statement is true or false.
CONTEXT: Each rompus is a wumpus. Every rompus is not opaque. Every jompus is a rompus. Every jompus is not sour. Vumpuses are jompuses [...] Polly is an impus.
STATEMENT: Polly is opaque.
ANSWER: true.

Figure 2: PrOntoQA-OOD Example

4 Experiments

4.1 Datasets

We evaluate our method on three diverse benchmarks for logical reasoning:

ProntoQA-OOD (Saparov et al., 2023) is a synthetic dataset for chain-of-thought first-order logic question-answering over abstract categories (see Figure 2). We consider two deduction settings: (1) tasks requiring only repeated applications of Modus Ponens; and (2) tasks involving all six supported deduction rules. For each setup, we partition the data by reasoning depth, selecting 600 examples (300 “true” and 300 “false”) for calibration and using the remaining 1,000+ examples for evaluation. To further test robustness, we adapt the scripts to vary the number of distractors (irrelevant context statements).

PARARULE Plus (Bao et al., 2022) is a dataset of true/false questions with fixed-depth reasoning. We use its test subset without modifications.

Extended-Multi-LogiEval builds on Multi-LogiEval (Patel et al., 2024). The original dataset is constrained by only 10 samples per logical scheme and an imbalanced answer distribution. We address these limitations by generating additional samples (100 per scheme) and enforcing a balanced 50/50

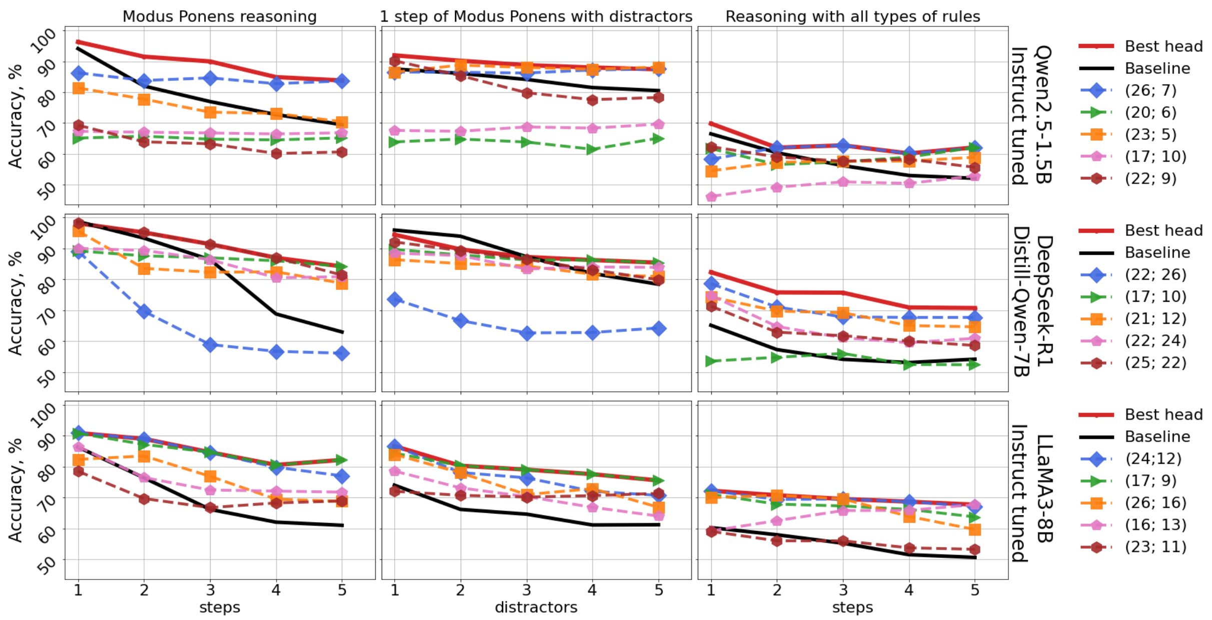

Figure 3: In-domain performamce on ProntoQA-OOD dataset. Best head was selected on calibration data for each case individually.

split for "yes" and "no" responses (treated as a_0 and a_1, respectively, for QK-score computation).

All experiments are performed in a zero-shot setup. Further details are provided in Appendices B and D.

4.2 Experimental Setup

All experiments were performed with frozen pretrained LLMs of size from 1.5B to 70B (for models larger than 7B, see Appendix E).

Our questions assume single-word answers; the standard approach in such setup is to select from a_0 and a_1 the option that was assigned the highest output probability by LLM, when the models are given the samples and asked to provide an answer. We refer to this method as the **Baseline**.

Our experiments were performed in two steps. First, for each explored setup from ProntoQA-OOD, we chose the best head in terms of QK-score on calibration set. Then we report the accuracy of QK-Scoring via the chosen head and the accuracy of Baseline on the evaluation subset.

Second, we selected five heads from those that achieved the top-10 performance in various ProntoQA-OOD setups (we aimed to choose heads that cover the higher number of setups). Then, we evaluated their performance on the PARARULE Plus and Extended-Multi-LogiEval datasets to asses the generalization capabilities of the QK-score informed head selection.

5 Results

5.1 In-Domain Evaluation

For each deduction setup in ProntoQA-OOD, we select the best-performing head on the calibration set (referred to as BEST HEAD) and evaluate its performance on the held-out evaluation set. Figure 3 shows that the BEST HEAD consistently outperforms the baseline across varying reasoning depths and under different numbers of distractors. In addition, we report the performance of five individual heads drawn from the top-five performers in each setup. These results indicate that the selected heads maintain stable performance regardless of the number of reasoning steps, thereby confirming that our QK-score reliably identifies transformer components that contribute to logical consistency. Results for three LLMs are presented here; further details are provided in Appendix E.)

5.2 Transfer Learning Evaluation

We further assessed the cross-dataset generalization of our approach by evaluating the QK-scoring accuracy of the heads selected in the above experiments on two additional datasets: PARARULE Plus and Extended-Multi-LogiEval. As reported in Table 1, three out of five selected heads achieve an accuracy exceeding the baseline by more than 10% on PARARULE Plus. In contrast, only two heads outperform the baseline on Extended-Multi-LogiEval, which may be attributed to differences

		PARARULE Plus Reasoning depth				Extended-Multi-LogiEval Reasoning depth			
	Head	**2**	**3**	**4**	**5**	**1**	**2**	**3**	**4**
Qwen2.5 1.5B, instruct	(26, 7)	0.6043	<u>0.6772</u>	0.6483	<u>0.7095</u>	0.4730	0.4663	<u>0.5163</u>	0.4413
	(20, 6)	0.5310	0.5844	0.5436	0.6097	**0.6069**	**0.6063**	**0.5804**	**0.5516**
	(23, 5)	0.5999	<u>0.6488</u>	0.6468	<u>0.6694</u>	0.5026	<u>0.5587</u>	0.4859	0.3840
	(17, 10)	0.4552	0.5569	0.4800	0.4549	0.5003	<u>0.5165</u>	0.5413	0.4943
	(22, 9)	**0.7095**	**0.7529**	**0.7719**	**0.8003**	0.2445	0.4531	0.4141	0.4527
Baseline	-	0.6240	0.6263	0.6521	0.6423	0.4996	0.4834	0.5011	0.5057
DeepSeek-R1 Distill-Qwen-7B	(22, 26)	**0.6736**	**0.7418**	<u>0.5495</u>	<u>0.6145</u>	0.5236	<u>0.5112</u>	<u>0.5402</u>	<u>0.5115</u>
	(22, 16)	0.3206	0.3422	0.2763	0.2388	0.4689	<u>0.5271</u>	<u>0.5185</u>	<u>0.5143</u>
	(21, 12)	0.4982	0.5203	0.4452	0.4740	**0.7682**	**0.5733**	**0.5989**	**0.6218**
	(22, 24)	<u>0.6523</u>	<u>0.6454</u>	**0.6149**	**0.6557**	<u>0.6572</u>	<u>0.5495</u>	<u>0.5771</u>	<u>0.5888</u>
	(25, 22)	<u>0.6227</u>	<u>0.6233</u>	<u>0.6145</u>	0.5357	0.5093	0.4768	0.4902	<u>0.5186</u>
Baseline	-	0.4969	0.5212	0.4523	0.4717	0.5153	0.4835	0.5010	0.5057

Table 1: Performance of QK-score on heads selected on ProntoQA-OOD in cross-domain evaluation. PARARULE Plus and Extended-Multi-LogiEval datasets are used. Best results are highlighted in **bold**. Those results that are better than baseline are <u>underlined</u>

in question format (binary "yes/no" vs. true/false evaluations).

Overall, these results demonstrate that our QK-score method identifies transformer heads capable of reliably assessing logical transitions, with robust performance observed both within the ProntoQA-OOD domain and in cross-dataset setups.

6 Analysis

Our findings suggest that QK-scores offer a distinctive lens on how LLMs process logical structure. Unlike raw attention weights, QK-scores are independent of positional encodings, thereby focusing purely on semantic alignment between the statement and candidate answers. Moreover, heads selected via QK-scores often outperform the model's final probabilities, confirming that essential reasoning signals can be sometimes obscured by later-stage processing (Kim et al., 2024; Lieberum et al., 2023).

We also observe that high QK-scoring heads consistently identify valid inferences even with distractors in the input. This stability indicates that such heads act as "verification anchors," largely unaffected by irrelevant context. Consequently, QK-scores may bolster both interpretability and robustness: by highlighting the heads that preserve logical consistency, they help clarify how multi-step reasoning unfolds within large language models.

7 Conclusion

We introduced a QK-score framework for uncovering attention heads that consistently capture logical validity. Our experiments show that in multi-step inferences with distractors, certain heads outperform the model's own final-layer predictions. Crucially, this single-pass procedure sidesteps the computational overhead of ablation-based methods and generalizes relatively well across datasets. By identifying attention heads that act as "reasoning checkpoints," our approach offers a tractable window into how these models process complex logical relationships. Future work may refine QK-scoring for specialized tasks, explore synergy with chain-of-thought prompting, and extend this analysis to other interpretability settings. Ultimately, bridging internal alignment signals with logical consistency represents a key step toward transparent and reliable language models.

8 Limitations

Application of our method requires a sufficiently large calibration dataset (at least ≈ 400 reasoning questions). This dataset must cover logical rules with more-or-less balanced number of examples and must be carefully debiased to prevent selection of attention heads that guess the answer from the form of the question.

We do not state that heads identified by our method are the only ones responsible for the logical inference within the model.

During the research for this short paper, we performed head selection only on two sets of deduction rules ("Modus Ponens" vs. "Modus Ponens, conjunction/disjunction introduction/elimination, and proof by contradiction"). It is quite possible, that the 'scope' of certain attention heads is limited to different sets of logical principles, and a more extensve experiments may be needed to explore those cases.

Finally, we considered only Instruct- and Chat-tuned models. Question of if and how capabilities for logic in attention heads change during the transition from base to fine-tuned version we leave for the future work.

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A Related Work

Here we discuss in more details the works related to logical reasoning and internal model interpretability.

Chain-of-Thought (CoT) and Logical Reasoning. CoT prompting has been widely adopted to improve reasoning in LLMs (Wei et al., 2022; Kojima et al., 2022; Yang et al., 2024; Seals and Shalin, 2023; Wan et al., 2024). These methods prompt models to produce intermediate steps in the reasoning process. However, despite the success in various tasks, they do not offer a measure to quantify the coherence of the reasoning transitions, leaving a gap that our QK-score method aims to fill.

Mechanistic Interpretability. A complementary line of research has focused on understanding transformer models through mechanistic interpretability. Work by Elhage et al. (2021) and Olah et al. (2020) has shown that transformer attention heads can be categorized into different functional phases, including knowledge recalling, latent reasoning, and expression preparation (Zheng et al., 2024). More recent studies have explored specific reasoning tasks: for example, Kim et al. (2024) examined syllogistic reasoning, showing that models learn content-independent reasoning mechanisms transferable across different logical schemes. Other investigations (Lieberum et al., 2023; Yu et al., 2024) have highlighted that certain heads can adversely affect the final decision by introducing latent biases. While ablation-based techniques (Zhang and Nanda, 2024; Todd et al., 2024; Yao et al., 2024) have been used to study these phenomena, they are often resource-intensive or limited to smaller models (Wang et al., 2023).

Benchmark Datasets. Several benchmarks have been designed to evaluate logical reasoning in LLMs. For example, LogicBench (Parmar et al., 2024) and Multi-LogiEval (Patel et al., 2024) test models on tasks such as truth table reasoning, logical entailment, and satisfiability. Datasets specifically tailored for chain-of-thought evaluation, such as PrOntoQA (Saparov and He, 2022), FOLIO (Han et al., 2022), and FLD (Morishita et al., 2024), demonstrate that unstructured intermediate reasoning steps can enhance performance.

Alternative Evaluation Strategies. Other lines of work have explored neuro-symbolic and data-driven methods to assess reasoning quality. Some approaches reformulate reasoning tasks into structured formats, while others propose direct evaluation metrics based on model outputs.

Summary. While previous work has substantially advanced our understanding of how LLMs reason and how their internal representations operate, there remains a need for efficient, scalable methods to assess logical consistency. Our method, which relies on the natural alignment between specific query and the last statement token key vectors, complements existing techniques and offers an efficient alternative to ablation-based analysis.

B ProntoQA-OOD: additional details

Questions in PrOntoQA-OOD are organized in a following way: given a set of axioms (context) it is required to prove a theorem (statement). For our study we reformulate them into answering if the statement is true or false given the context.

We used scripts publsihed by the authors of the

Deduction Rule	Number of hops				
	1	2	3	4	5
M. P.	6,144	6,204	6,364	6,288	6,176
M. P. + distr.	6,268	-	-	-	-
Comp.	2,752	2,764	2,884	2,956	2,892

Table 2: Size of ProntoQA-OOD evaluation set that was used in different setups

dataset to generate the data. We used following command line flags:

- For Modus Ponense only:
  `--ordering random --distractors none --deduction-rule ModusPonens --proofs-only --ontology fictional --min-hops 1 --max-hops 5`

- For composition of deduction rules:
  `--ordering random --deduction-rule Composed --proofs-only --distractors none --ontology fictional --min-hops 1 –max-hops 5`

We also modified the original scripts to make possible variating the number of distractors in prompts, and generated data for $1-$hop questions on Modus Ponense with $1, 5$ distractors respectively.

In our experiments we used training data from *in context examples*. For each setup (deduction rule+number of hops + number of distractors) scripts yielded $4,000$ samples. To exclude possible biases, in each case we select equal number of questions where the statement is given in positive (`X is Y.`) and negative (`X is not Y.`) forms and for each sample we also generate its counterpart where its statement and ground-truth answer are negated (thus balancing the classes). Depending on the setup, this resulted in $2,600 - 7,000$ questions, out of which 600 ($300/300$ with answers "true" and "false" respectively) were taken for calibration and the rest were used as evaluation set (see Table 2). We ensured that no pair axioms+theorem is included in both subsets.

C Pararule-plus: additional details

Here we demonstrate the example from this dataset:

CONTEXT: Harry is strong. Harry is big. Harry is high. Anne is thin. Anne is little. Gary is smart. Gary is quiet. Gary is kind. Fiona is poor. Fiona is rough. Fiona is sad. Strong people are smart. If someone is thin and little then they are short. If someone is poor and rough then they are bad. If someone is smart and quiet then they are nice. All short people are small. All smart people are quiet. All nice people are wealthy. All bad people are dull.
STATEMENT: Harry is quiet.
ANSWER: true

Figure 4: PARARULE-PLUS prompt example of reasoning, depth 2

On PARARULE Plus, three out of five heads, selected on the ProntoQA-OOD dataset, reach accuracy higher than the baseline, in most cases by more than 10%. Interestingly, head `(22, 16)` from DeepSeek-R1 consistently yields an accuracy below 0.35 on PARARULE Plus, suggesting that its QK-score distinguishes correct from incorrect logical implications but in a reversed manner. Similar effect also occurs in some setups on other heads.

D Extended Multi-Logi-Eval: additional details

Here we demonstrate the example from this dataset and how it was prompted:

Use provided Context to answer the Question. Print 'yes' or 'no' only.
CONTEXT: If a person uses a fishing rod, they catch fish. Michael uses a fishing rod.
QUESTION: Does Michael catch fish?
ANSWER: yes.

Figure 5: Multi-LogiEval Modus Ponens Example of reasoning depth 1

While the original dataset Multi-Logi-Eval consisted of three types of logic, namely First-Order Logic, Nonmonotonic Logic and Propositional Logic, we extended only a part with first-order logic. We used GPT-4o to generate more samples for each scheme. Table 3 compares the statistics of generated dataset with the statistics of the original (Multi-LogiEval dataset):

Dataset	Depth 1	2	3	4
Multi-LogiEval (FOL)	130	105	135	120
Extended Multi-LogiEval	1300	700	900	700

Table 3: Statistics of generated Extended-MultiLogiEval dataset.

We refer to original paper (Patel et al., 2024) for the detailed description of logical schemes, and we did not add any new schemes or changed them in any way on all levels of reasoning. Every scheme of reasoning depth k consists of k atomic rules from reasoning depth 1, see them in Table 4.

E Extended results of in-domain evaluation on ProntoQA-OOD

Tables 5 and 6 provide the in-domain numerical evaluation results on ProntoQA-OOD for BEST HEAD and BASELINE methods calculated for various LLMs.

Rule	First-order Logic
MP	$(\forall x(p(x) \rightarrow q(x)) \land p(a)) \vdash q(a)$
MT	$(\forall x(p(x) \rightarrow q(x)) \land \neg q(a)) \vdash \neg p(a)$
HS	$(\forall x((p(x) \rightarrow q(x)) \land (q(x) \rightarrow r(x))) \vdash (p(a) \rightarrow r(a))$
DS	$(\forall x(p(x) \lor q(x)) \land \neg p(a)) \vdash q(a)$
CD	$(\forall x((p(x) \rightarrow q(x)) \land (r(x) \rightarrow s(x))) \land (p(a) \lor r(a))) \vdash (q(a) \lor s(a))$
DD	$(\forall x((p(x) \rightarrow q(x)) \land (r(x) \rightarrow s(x))) \land (\neg q(a) \lor \neg s(a))) \vdash (\neg p(a) \lor \neg r(a))$
BD	$(\forall x((p(x) \rightarrow q(x)) \land (r(x) \rightarrow s(x))) \land (p(a) \lor \neg s(a))) \vdash (q(a) \lor \neg r(a))$
CT	$\forall x(p(x) \lor q(x)) \dashv\vdash \forall x(q(x) \lor p(x))$
DMT	$\neg \forall x(p(x) \land q(x)) \dashv\vdash \exists x(\neg p(x) \lor \neg q(x))$
CO	$\forall x((p(x) \rightarrow q(x)) \land (p(x) \rightarrow r(x))) \vdash \forall x(p(x) \rightarrow (q(x) \land r(x)))$
IM	$\forall x(p(x) \rightarrow (q(x) \rightarrow r(x))) \dashv\vdash \forall x((p(x) \land q(x)) \rightarrow r(x))$
EG	$p(a) \vdash \exists x(p(x))$
UI	$\forall x(p(x)) \vdash p(a)$

Table 4: Inference rules that establish the relationship between premises and their corresponding conclusions in Extended Multi-LogiEval. The schemes are MP: Modus Ponens, MT: Modus Tollens, HS: Hypothetical Syllogism, DS: Disjunctive Syllogism, CD: Constructive Dilemma, DD: Destructive Dilemma, BD: Bidirectional Dilemma, CT: Commutation, DMT: De Morgan's Theorem, CO: Composition, IM: Importation, EG: Existential Generalization, UI: Universal Instantiation

		Modus ponens only					Composition of rules				
Reasoning depth:		**1**	**2**	**3**	**4**	**5**	**1**	**2**	**3**	**4**	**5**
LLaMA2 7B Chat	QK	**0.8889**	**0.8427**	**0.8474**	**0.8524**	**0.8444**	**0.7474**	**0.7847**	**0.7237**	**0.7013**	**0.7004**
	Baseline	0.6772	0.6349	0.6506	0.6197	0.6072	0.5678	0.5414	0.5427	0.5232	0.5346
LLaMA2 13B Chat	QK	**0.9778**	**0.9637**	**0.9283**	**0.9187**	**0.9107**	**0.7584**	**0.7134**	**0.7044**	**0.6777**	**0.6964**
	Baseline	0.8944	0.7784	0.7527	0.7385	0.7414	0.6556	0.4944	0.4932	0.4751	0.4786
LLaMA3 8B Instruct	QK	**0.9090**	**0.8899**	**0.8465**	**0.8054**	**0.8217**	**0.7219**	**0.7073**	**0.6951**	**0.6863**	**0.6767**
	Baseline	0.8632	0.7636	0.6613	0.6199	0.6097	0.6023	0.5791	0.5517	0.5145	0.5057
LLaMA3.1 8B Instruct	QK	**0.9881**	0.9577	0.9192	0.8799	0.8676	**0.8084**	**0.7184**	**0.7188**	**0.7239**	**0.7052**
	Baseline	0.9843	**0.9785**	**0.9473**	**0.9035**	**0.8775**	0.6848	0.6206	0.5847	0.5888	0.5731
LLaMA3 70B Instruct	QK	**1.0000**	**0.9999**	0.9963	**0.9981**	**0.9960**	0.**9890**	**0.9214**	**0.8799**	**0.8617**	**0.8353**
	Baseline	0.9993	**0.9999**	**0.9981**	0.9868	0.9739	0.9805	0.8767	0.8324	0.8171	0.7948
LLaMA3.1 70B Instruct	QK	0.9987	0.9993	**0.9993**	**0.9968**	**0.9942**	0.9890	**0.9144**	**0.9073**	**0.8762**	**0.8631**
	Baseline	**1.0000**	**0.9999**	0.9967	0.9904	0.9762	**0.9922**	0.8912	0.8685	0.8348	0.8246
Qwen-2.5 1.5B Instruct	QK	**0.9627**	**0.9149**	**0.8995**	**0.8487**	**0.8379**	**0.6981**	**0.6201**	**0.6275**	**0.6135**	**0.6209**
	Baseline	0.9410	0.8197	0.7694	0.7273	0.6946	0.6648	0.6036	0.5617	0.5299	0.5207
Qwen-2.5 14B Instruct	QK	**0.9988**	**0.9966**	**0.9869**	**0.9791**	**0.9683**	**0.9724**	**0.8813**	**0.8313**	**0.8332**	**0.8397**
	Baseline	0.9660	0.9656	0.9388	0.9146	0.9060	0.9050	0.8374	0.7720	0.7323	0.7451
Qwen-2.5 32B Instruct	QK	**1.0000**	**0.9996**	**0.9947**	**0.9929**	**0.9916**	**0.9543**	**0.9257**	**0.8598**	**0.8340**	**0.8104**
	Baseline	0.9988	0.9975	0.9783	0.9203	0.8502	0.9274	0.8623	0.8264	0.7982	0.7710
DeepSeek R1-Distill-Qwen-1.5B	QK	**0.9149**	**0.8039**	**0.8995**	**0.8487**	**0.8379**	**0.6718**	**0.6393**	**0.5976**	**0.5825**	**0.5894**
	Baseline	0.5361	0.5103	0.5053	0.5011	0.5020	0.4873	0.4980	0.4959	0.4971	0.4985
DeepSeek R1-Distill-Qwen-7B	QK	0.9816	**0.9519**	**0.9141**	**0.8701**	**0.8418**	**0.8233**	**0.7577**	**0.7568**	**0.7090**	**0.7071**
	Baseline	**0.9873**	0.9336	0.8638	0.6879	0.6295	0.6512	0.5723	0.5408	0.5304	0.5413

Table 5: Comparison of models with different reasoning depths on ProntoQA-OOD. Best results are highlighted in bold.

		Distractors added				
		1	**2**	**3**	**4**	**5**
LLaMA2	QK	**0.8725**	**0.8654**	**0.8595**	**0.8487**	**0.8498**
7B Chat	Baseline	0.6577	0.6583	0.6422	0.6280	0.6248
LLaMA2	QK	**0.9627**	**0.8972**	**0.8981**	**0.8920**	**0.8552**
13B Chat	Baseline	0.7959	0.7348	0.7160	0.7005	0.6677
LLaMA3	QK	**0.8665**	**0.8028**	**0.7902**	**0.7753**	**0.7551**
8B Instruct	Baseline	0.8632	0.7636	0.6613	0.6199	0.6097
LLaMA3.1	QK	0.9527	0.8577	**0.8170**	**0.7895**	**0.7699**
8B Instruct	Baseline	**0.9629**	**0.8691**	0.7830	0.7212	0.6672
LLaMA3	QK	**0.9967**	**0.9947**	**0.9897**	**0.9849**	**0.9845**
70B Instruct	Baseline	0.9961	0.9882	0.9463	0.9106	0.8926
LLaMA3.1	QK	**0.9971**	**0.9954**	**0.9853**	**0.9672**	**0.9613**
70B Instruct	Baseline	0.9933	0.9918	0.9582	0.9176	0.8916
Qwen-2.5	QK	**0.9627**	**0.9149**	**0.8995**	**0.8487**	**0.8379**
1.5B Instruct	Baseline	0.9410	0.8197	0.7694	0.7273	0.6946
Qwen-2.5	QK	**0.9955**	**0.9823**	**0.9556**	**0.9362**	**0.9320**
14B Instruct	Baseline	0.9687	0.9528	0.8896	0.8336	0.8130
Qwen-2.5	QK	**0.9997**	**0.9973**	**0.9794**	**0.9546**	**0.9429**
32B Instruct	Baseline	0.9990	0.9931	0.9253	0.8042	0.7522
Deepseek	QK	**0.8290**	**0.7650**	**0.7485**	**0.7574**	**0.7501**
R1-Distill-Qwen-1.5B	Baseline	0.5361	0.5096	0.5098	0.5016	0.5012
Deepseek	QK	0.9448	0.8972	0.8725	**0.8625**	**0.8540**
R1-Distill-Qwen-7B	Baseline	**0.9597**	**0.9397**	**0.8736**	0.8191	0.7840

Table 6: Effect of distractors added to the prompt on ProntoQA-OOD. Only Modus Ponens inference.